

Narrative Project Proposal: Competition Lunar Construction Robot, Matthew Geisel

Purpose, Background, and Significance

NASA's Artemis Mission intends to further the mission of human space exploration by developing a permanent settlement on the moon to prepare for human settlement on Mars. NASA has created numerous competitions to facilitate idea generation and research into various problems that Artemis will face. The purpose of this project is to conduct research to assist NASA by competing in one such competition.

The problem this project will address has to do with lunar regolith (moon dust). Lunar regolith is a fine, sharp powder that is comparable to tiny shards of glass. When a rocket takes off, the force used to propel it jettisons a fine powder of this regolith all around it (much like the plumes of smoke seen in an Earth rocket launch). Unlike Earth, the Moon has no air resistance and only 16.6% of Earth's gravity, meaning the particles keep their velocity and take longer to fall to the surface. These fast-moving tiny glass-like particles can rip through electronics and habitations, endangering the Lunar Colony.

NASA scientists have determined that an optimal solution to this problem is the construction of Lunar Berms. Berms are ridges of sediment that are used on Earth for drainage and wave protection (like the ridges of sand and grass at the back of a beach). On the moon, these berms can be built around rocket launch sites to prevent regolith from damaging sensitive equipment. The purpose of this project is to automate this berm construction, creating a robot that effectively traverses lunar terrain and constructs berms out of lunar regolith. This robot is intended to compete in Iowa in an independent competition in 2025 and in Cape Canaveral in the NASA competition in 2026.

Objectives and Methods

The objective of this project is to create a robot that can autonomously traverse lunar regolith, excavate lunar regolith, and construct mounds (simulating berms) with it. This robot will compete on a 6.88 meter by 5 meter field filled with simulated lunar regolith. The robot will have to autonomously navigate to the designated Excavation Zone while dodging boulders and craters, excavate regolith, move to the construction zone, and construct a berm with the simulated regolith. The efficacy will be measured by berm volume with respect to robot mass and power consumption.

Driving over lunar regolith is similar to driving over sand. This means that a normal wheel will not generate enough traction to effectively move the robot. Therefore, a wheel with grousers will be used to get the required traction to move. However, these wheels will still be non-ideal when it comes to driving, and turning will be difficult. Therefore, a *swerve drive* will be implemented, which allows for wheels to turn independently, meaning the robot can move forward, backwards, and sideways.

Construction will be accomplished using a belt with scooping attachments that will scoop sediment and transport it up the belt. This belt will lift the sediment into an attachment at the back of the robot, which will hold the sediment (much like a dump truck). A door will open at the bottom of this attachment, allowing the sand to fall onto the ground. The excavator will be stowed on top of the robot using a four-bar mechanism.

Due to the low traction of the regolith, using "dead reckoning" (determining position using wheel rotations) will be very inaccurate. Therefore, *beacons* will be used to determine the position of the robot. UWB (Ultra Wideband) modules can send messages between each other and determine position (this is the technology behind Air Tags). This technology can be extended to this use case, as static beacons containing UWB modules can be planted in the regolith and their positions are fixed. UWB modules with

an accuracy of 10 cm have already been specified, and four of these (three beacons and one robot module) will be used. An algorithm will be used to determine the most probable location of the robot based on these measurements.

Obstacle avoidance will be accomplished using a LiDAR (Light Detection and Ranging) sensor. This sends a light pulse and measures the time for it to return. A low-cost LiDAR sensor has been specified that can measure the distances in two dimensions. This will be added to a motor that moves up and down, performing a sweep of the floor in front of it. If an irregularity is detected, the robot will avoid it.

Numerous challenges may arise in undertaking this project in both the mechanical and software aspects. Mechanically, swerve drives are very complex systems that have many more failure points than traditional drive systems. Therefore, the system will likely experience failures in testing that will have to be addressed. By creating a schedule that allows for testing time, these failures will be addressed before the competition. The construction system will be tested on Revere beach and other nearby beaches before competition to determine failure points.

A principal software challenge is intermittent bugs. Critical bugs that create failures are easier to detect and solve for, as they will come up in development and testing. However, intermittent bugs (ones that only occur once in a while) are much more difficult to find before competition. In order to address this challenge, ROS2 (Robot Operating System) will be used to control aspects of the robot. This creates abstract “nodes” which each perform their own function. Bugs in nodes can be addressed individually and debugged separately, reducing debug time

Another possible challenge is inaccuracies in sensors. For example, UWB sensors have an accuracy of 10cm. This means that a distance of 1m could be read anywhere between 0.9 - 1.1 m. In order to account for this error, beacons will be placed as far from each other as possible and probability fields will be used. LiDAR and UWB functionality will be tested off the robot to ensure they are working.

This project is being conducted under the supervision of SEDS (Students for the Exploration and Development of Space). The project intends to use the 3D printers at the basement of Richards, as well as the machine shop adjacent to it (with a Lathe, CNC Mill, and Tormach) for machining steel and aluminum. The EXP makerspace will be used for its Resin 3D printer and Laser Cutter for machining Acrylic. This project will also use all other resources of SEDS, such as advice from other SEDS projects such as NURover.

Matthew Geisel is planning the project to be completed April 27, 2025 (end of Spring Semester). The drive system design will be finalized by February 1, so it can be manufactured by March 1. While the drive system is being constructed, the construction system will be designed (completed by March 1) and then constructed by April 1. These systems will then be tested until the competition.

Before the drive system is completed, ROS2 will be implemented on the Raspberry Pi and the trilateration algorithm will be implemented (by February 1). Then, the Lidar object detector will be created and the user control system will be developed (by March 1). Then, the swerve drive will be programmed and tested by April 1, and the construction system will be programmed and tested by April 14. The period between April 14 and April 27 will be reserved for testing.

Outcomes, Evaluation, and Dissemination

This project has four critical milestones. The first milestone is a complete CAD assembly of the entire system, including 3D printed parts, electronic components, manufactured parts, and all other hardware components. This will be considered done after it has been reviewed and approved by SEDS

eboard and/or the NURover mechanical team. The second milestone is to have the entire robot constructed per the confines of the CAD. This will be considered complete when the robot is rigid (parts do not move unless desired) and moves as desired. The third milestone is to complete the autonomous code. This will be considered complete when the robot can autonomously move from one point to another within the designated area to an accuracy of 50 cm while avoiding obstacles. The fourth milestone is total robot control (drive, excavation, and construction). This will be considered complete when a user can easily drive, excavate, and construct berms without looking at the robot.

The expected outcomes of this project are a fully functional robot that can drive on simulated lunar regolith, autonomously navigate to a specified destination while avoiding obstacles, excavate simulated lunar regolith, and construct berms out of lunar regolith. This project is a research project that will be led by Matthew Geisel and will be completed by April 27, 2025. This project will be shared first with other teams at the Iowa State University competition on May 15 and then with NASA in the Lunabotics competition in 2026. Hopefully, NASA will take some of the ideas from this research to develop the Artemis Program and establish a lunar colony.

About the Learner

I joined SEDS and the Northeastern University Mars Rover Team my first semester at Northeastern (fall of 2022). Since joining, I have been a member of the Mechanical team, responsible for designing components such as the arm and drivetrain. I am pursuing this project so I can develop a new system and get more experience than I can get from working on iterative design. I then progressed up the SEDS leadership, becoming first Events Coordinator and then Finance Chair, developing a custom finance system for managing purchases. I am a Mechanical and Electrical Engineering major with experience in CAD, PCB design, firmware development, and webdev, and I believe I am capable of leading and completing this project and competing in the spring.

Being a third year Mechanical and Electrical Engineering student, I have a wide variety of both class, club (EWB, NURover, SEDS), research (UNLab), and co-op (Notch Inc.) experience that has given me the experience required to be at the summit of my college career. Through these experiences, I have developed the skills required to manage a team and complete an engineering project of this scale. The SEDS president and I found the competition, and I worked to develop a plan and how to tackle it. I will have full responsibility for making decisions and I will lead a team to help achieve this goal. I will seek guidance with my faculty mentor through email if I experience any complex challenges I am unsure of how to face.

After college, I want to either found a consumer robotics startup or work in an engineering or managerial position at an engineering firm. This will allow me to advance my engineering and managerial skills, and learn what it's like to start a project from the ground up.